

Forecasting Clinical-Trial Drug Demand under Uncertain Enrolment and Consumption

A compound-distribution model, cross-validated by three complementary methods

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Abstract

This note forecasts the total drug a clinical-trial depot must stock under uncertain enrolment and consumption, and validates the forecast by *independent, complementary methods*. Total demand is modelled as a compound distribution: an over-dispersed Negative-Binomial patient count and a per-patient kit severity derived from the dosing schedule and the dropout hazard. Its distribution is computed *exactly* by FFT convolution of the compound generating function. The exact reference is cross-checked against a Monte-Carlo simulation, closed-form moments, a known-answer case, and an independent implementation in the `aggregate` library; the methods agree on the 99th-percentile demand to the kit. On a real GLP-1 cohort, typical demand (median 416 kits) is dwarfed by the coverage requirement ($q_{99} \approx 97,000$ kits per depot) — a roughly 230-fold gap driven by heavy-tailed enrolment — which quantifies the overage a service guarantee actually costs. Where the methods first disagreed, the discrepancy was traced to FFT aliasing and resolved by sizing the lattice to the distribution. The contribution is not methodological novelty but a reproducible computational framework, built from public trial registries, in which a demand forecast is certified rather than asserted.

1. The problem

A clinical trial must place drug at its supply depots *before* it knows how much will be needed. Demand is uncertain on two fronts at once: how many patients will enrol at the sites a depot serves, and how much drug each consumes before completing or dropping out. Ship too little and a patient misses a dose — a protocol deviation; ship too much and costly product expires. For an expensive biologic, both errors are measured in real money, so the buffer stock — the *overage* — is a genuine budget decision. The question is deliberately narrow: **how much should a depot stock so it almost never runs short?** The answer is not one number but a distribution of possible total demand, and, above all, its upper tail — the region a service guarantee is priced against.

The organising principle of this note is that such a forecast is only trustworthy if the same distribution, computed by *independent methods*, agrees. A single simulator confirms that code runs, not that the numbers are right; the tail that sizes overage is precisely where a simulation is least reliable. The demand distribution is therefore computed several ways — an exact FFT reference, a Monte-Carlo simulation, and the field-standard `aggregate` library — that must reconcile.

2. The model

2.1 Total demand as a compound sum

Let N be the random number of patients a depot serves over the planning horizon and X_i the random number of kits patient i consumes. Total demand is the compound sum

$$S = \sum_{i=1}^N X_i, \quad S = 0 \text{ if } N = 0.$$

The patient count N is over-dispersed. If every trial recruited at the same fixed rate, the count would be Poisson, whose variance equals its mean. Real enrolment is far more variable — some trials recruit much faster than others — so it is *over-dispersed* and modelled as a Negative Binomial. The mechanism is a mixed-Poisson one: let the Poisson rate itself be random, drawn from a Gamma distribution of mean one; averaging the Poisson over the Gamma-drawn rates yields exactly a Negative Binomial. This is the standard identity *Poisson rate mixed by a Gamma = Negative Binomial*. By the law of total variance,

$$\text{Var}(N) = \underbrace{\mathbb{E}[\text{Var}(N \mid \Lambda)]}_{\text{Poisson noise given the rate}} + \underbrace{\text{Var}(\mathbb{E}[N \mid \Lambda])}_{\text{spread of the rate itself}} = m + c^2 m^2,$$

where $m = \mathbb{E}[N]$ and c is the coefficient of variation of the Gamma mixing distribution. The second term is the over-dispersion; it grows with m^2 and, at this cohort's scale, dominates — producing the heavy enrolment tail that drives the demand tail. Fitting by moments to the depot-level counts $m, v = \text{Var}(N)$ gives $p = m/v$ and $r = m^2/(v - m)$ for $N \sim \text{NegBin}(r, p)$.

Per-patient consumption X is constructed from the regimen. A patient receives one kit per dosing cycle of length Δ months and survives each cycle with probability $q = e^{-\lambda\Delta}$, where the dropout hazard $\lambda = -\ln(c_{\text{cmp}})/T$ is recovered from the observed completion fraction c_{cmp} over a treatment period T . The number of administrations before dropout, capped at the horizon, is truncated geometric, and $X = (\text{dose per administration}) \times (\text{administrations})$.

2.2 The exact solution by FFT

The probability generating function (pgf) of a compound sum factorises,

$$G_S(z) = \mathbb{E}[z^S] = G_N(G_X(z)),$$

with count pgf $G_N^{\text{Poisson}}(z) = e^{\lambda(z-1)}$ or $G_N^{\text{NegBin}}(z) = (p/(1 - (1-p)z))^r$. Evaluating a pgf on the unit circle is the discrete Fourier transform, so for a severity pmf $\mathbf{f}_X$ on an integer kit lattice of length M the aggregate pmf is one forward and one inverse FFT:

$$\mathbf{f}_S = \mathcal{F}^{-1} \left[G_N(\mathcal{F}[\mathbf{f}_X]) \right].$$

This is the collective-risk / aggregate-loss machinery of actuarial science, applied to clinical supply; it is exact up to lattice discretisation. The same engine serves the identical role in the author's longevity-risk work, where the compound-Poisson FFT valuation is reused across a grid of structural mortality scenarios.

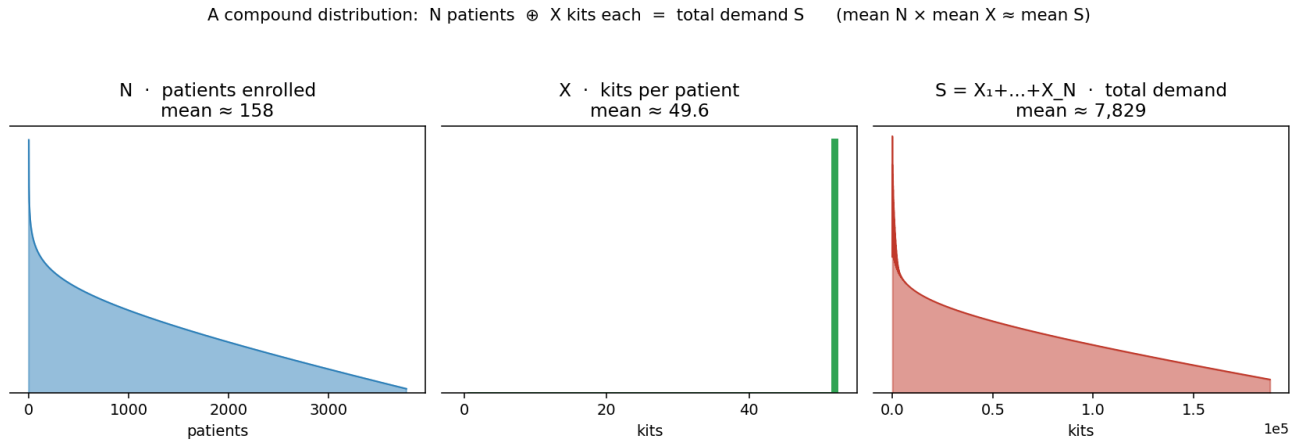


Figure 1: The compound structure: a random patient count N , a per-patient kit draw X , and their pooled total S , with $\mathbb{E}[N] \times \mathbb{E}[X] \approx \mathbb{E}[S]$.

3. Validation

The exact FFT reference is cross-checked four ways. (i) A *known-answer* case: with degenerate severity $X \equiv 1$ the sum is the count, $S = N$, and the FFT reproduces the exact Poisson/NegBin pmf to machine precision ($\sim 10^{-16}$). (ii) *Moment identities*: $\mathbb{E}[S] = \mathbb{E}[N]\mathbb{E}[X]$ and $\text{Var}(S) = \mathbb{E}[N]\text{Var}(X) + \text{Var}(N)\mathbb{E}[X]^2$, reproduced by both the FFT and the simulation. (iii) A *Monte-Carlo* simulation — draw N , draw N severities, sum, repeat — agreeing across the distribution and, crucially, at the overage quantile.

The agreement is not a coincidence of one sample: as the path count grows, the Monte-Carlo tail quantile *converges* onto the exact FFT value, confirming that the FFT is the stable target the simulation homes in on rather than a chance match. Roughly 10^4 – 10^5 paths are needed to pin the overage tail to within a per-cent.

(iv) A second *exact* method from an independent codebase: Mildenhall's `aggregate` library, fed the same discrete kit severity and the same count law. The translation is where the Gamma returns — `aggregate` takes the

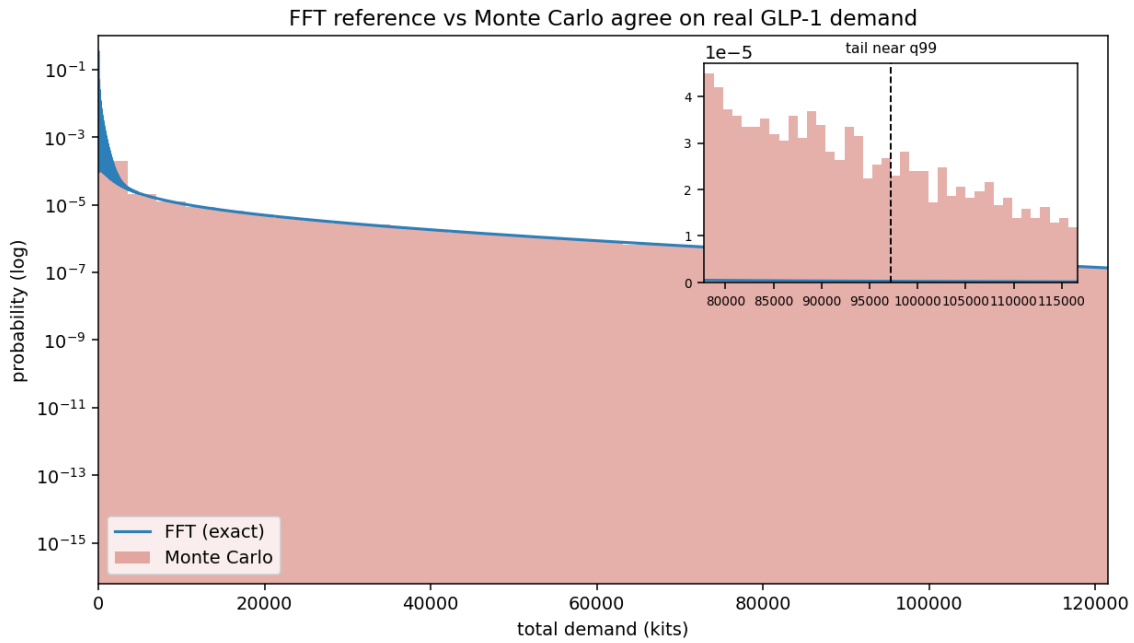


Figure 2: The exact FFT reference and the Monte-Carlo simulation agree on the real GLP-1 demand distribution, including the q_{99} tail (inset).

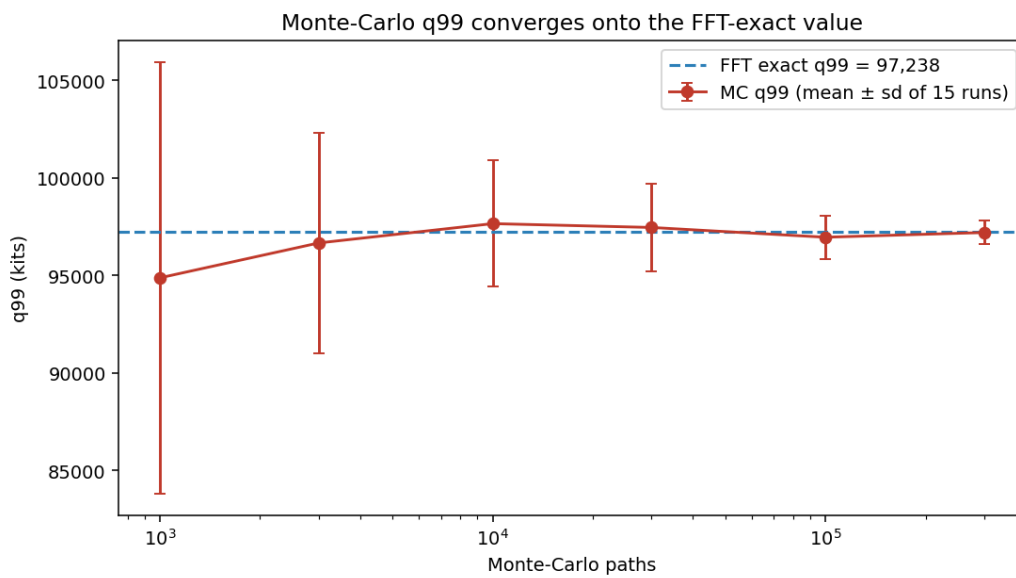


Figure 3: Monte-Carlo q_{99} converges onto the exact FFT value as the number of simulation paths grows; the error bars are the spread over fifteen independent runs.

mixing description mixed gamma c , not (r, p) , so one sets $c = \sqrt{v - m}/m \approx 2.50$ to reproduce the identical Negative Binomial. The four methods land on the same distribution:

Method	q_{50}	q_{95}	q_{99}
From-scratch FFT	416	42,522	97,238
Monte Carlo (200k paths)	416	42,265	96,167
aggregate (library)	416	42,522	97,238

The Monte-Carlo q_{99} sits slightly low because the far quantile is where simulation is noisiest; it converges onto the exact value as paths grow. The exact FFT and the reference library agree exactly.

The aliasing finding. The methods did not agree at first: on the real cohort the FFT read q_{99} about 12% low. The cause was aliasing — the heavy enrolment tail carried demand past the end of the FFT lattice, which wrapped around and truncated exactly the tail that sizes overage. Sizing the lattice to the distribution — growing M until the wrapped mass is negligible — resolved it, after which all methods agree and the aliased mass falls below 10^{-11} . The value of the cross-check is precisely this: it surfaced a real numerical failure that any single method would have hidden.

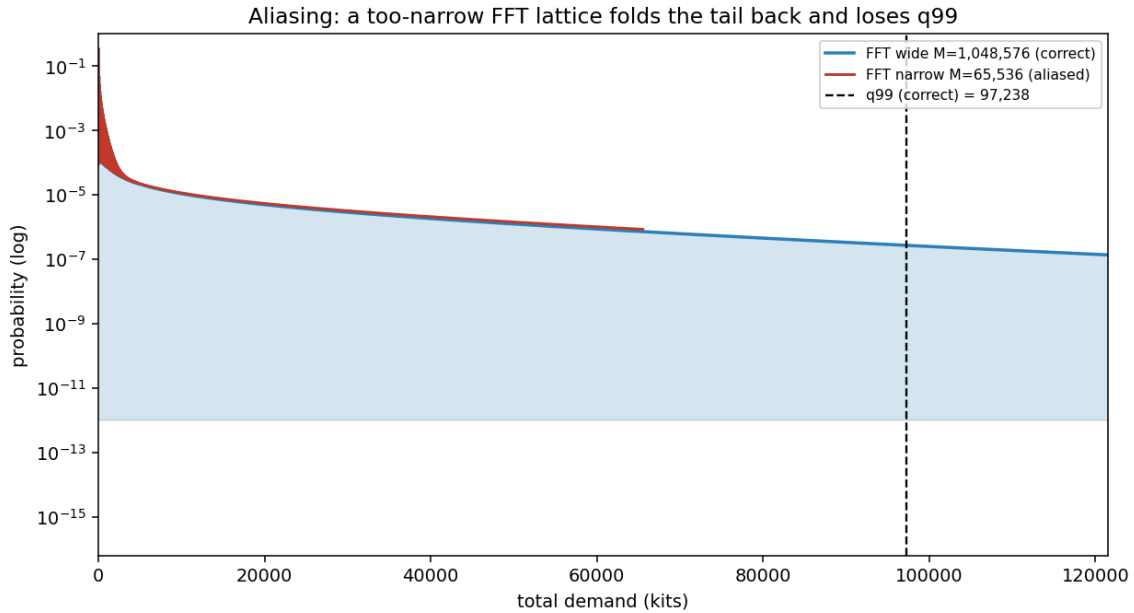


Figure 4: A too-narrow FFT lattice folds the demand tail back on itself and loses q_{99} ; sizing the lattice to the distribution restores it.

4. Reading the results

For a real GLP-1 cohort (weekly injectable, ~ 72 -week treatment), a depot serving five sites over a twelve-month horizon:

Quantile	Total demand (kits)	Meaning
q_{50}	416	typical demand
q_{95}	42,522	covered 95% of the time
q_{99}	97,238	covered 99% of the time

The striking feature is the roughly 230-fold gap between the median and q_{99} . Typical demand is small, but rare high-enrolment scenarios are enormous, so a service guarantee costs far more than the average suggests. This

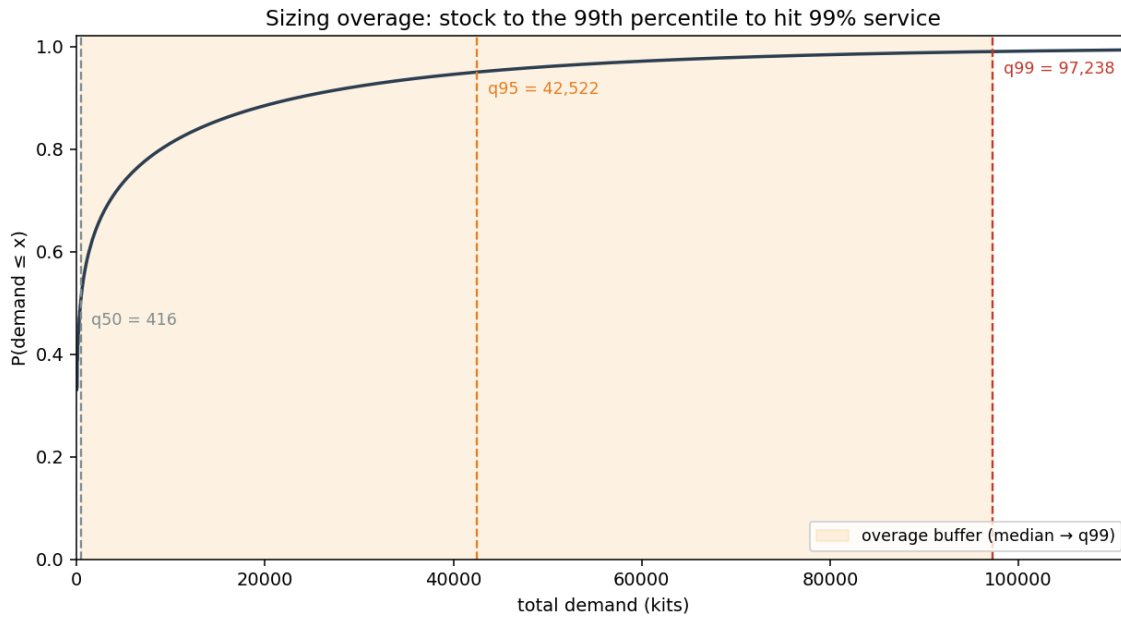


Figure 5: Sizing overage: stock to the 99th percentile to hit 99% service; the shaded band is the buffer above typical demand.

is not waste — it is the price of covering the tail — and the cumulative distribution above is the instrument for the actual budget decision: choosing a *service level*. Stocking to q_{95} rather than q_{99} more than halves the buffer, at the cost of running short 5% of the time instead of 1%. The heavy tail is a property of this cohort’s enrolment variability, which the model exposes rather than creates; a planner working from a single average would never see it.

5. Scope: where the model applies — and where it stops

The compound structure requires *uncertainty in the patient count* and *supply committed ahead of enrolment*. It fits small molecules and oral drugs, and — the high-value case — monoclonal antibodies and other biologics dispensed from depot stock over a trial, where high unit cost and forecast-ahead supply coincide. It does **not** fit bespoke, per-patient therapies such as CAR-T, where each dose is manufactured for an already-identified patient: there is no depot to pre-stock and no count to forecast, and supply is a scheduling problem against known individuals rather than a distribution. Stating the boundary is part of the method — the model earns its keep precisely where drug cost is highest and supply is forecast against an uncertain count.

6. Scenario knobs

Several inputs are planning *choices*, not facts from the data; they define the scenario, and the headline figures scale with them. Two others are pinned by the regimen.

Knob	Value	Kind	Meaning
K_SITES	5	choice	Sites feeding one depot; more sites → more demand.
HORIZON_M	12	choice	Months of enrolment planned ahead.
DOSE_PER_ADMIN_KITS	1	choice	Kits per administration (a titrating regimen would vary it).

Knob	Value	Kind	Meaning
T_TREAT_M	16.6	data	Treatment period (~ 72 wk); sets the dropout hazard λ .
ADMIN_INTERVAL_M	0.23	data	Dosing interval (weekly); sets kits accrued per patient.

HORIZON_M and T_TREAT_M are distinct clocks: the first is how far ahead the *depot* plans (scaling the count N), the second how long a *patient* stays on the drug (scaling consumption X). Because K_SITES and HORIZON_M scale enrolment, the headline $q_{99} \approx 97,000$ kits is *per depot, five sites, twelve-month horizon*; the transferable insight is the shape — a small median beside an enormous tail — not the specific figure.

7. Reproducibility

Every parameter traces to real completed Phase-3 trials on ClinicalTrials.gov (by NCT id), curated to chronic parallel-group designs, with dosing cross-checked against DailyMed. The simulation then runs from those parameters, so the ground truth is known and the study reproduces without re-fetching: *real public parameters, synthetic generating layer*. The full implementation, figures and tests are in the accompanying notebook, `clinical_supply_validation.ipynb`.